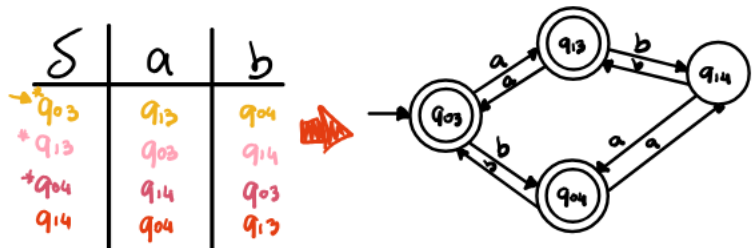
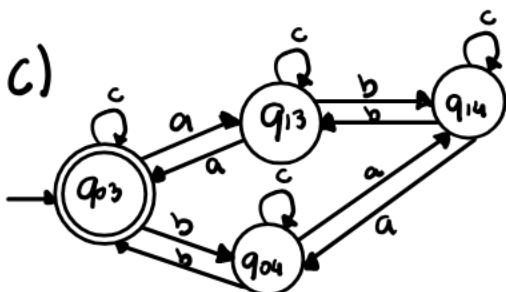
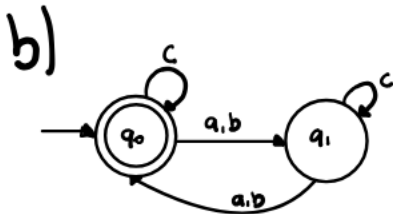
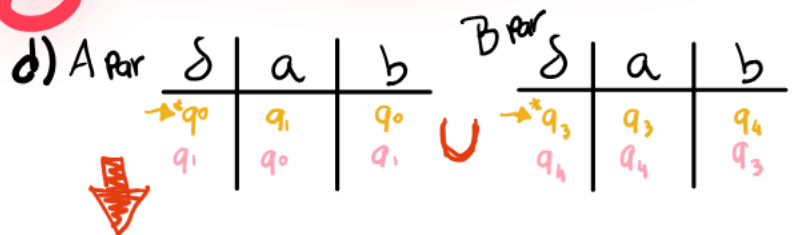
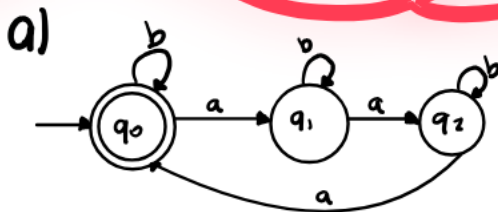


Lista de Exercícios:
Projeto de Autômatos Finitos

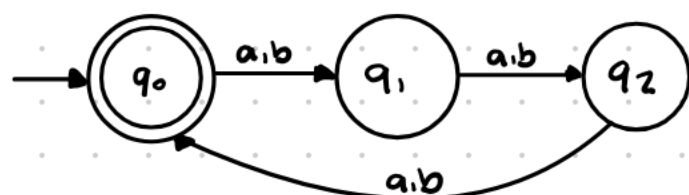
Linguagens Formais e Compiladores
Profª. Jerusa Marchi

1. Construa Autômatos Finitos Determinísticos que reconheçam as seguintes Linguagens:

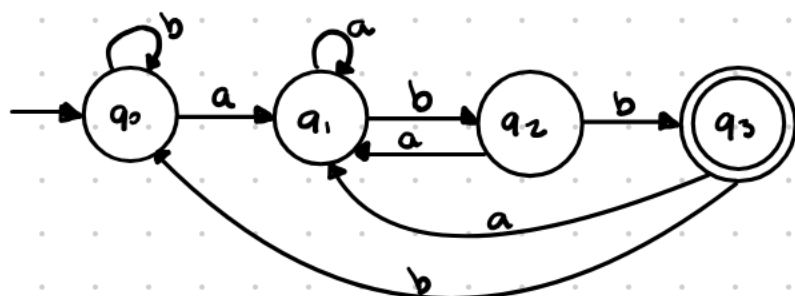
- (a) $L = \{w \mid w \in \{a, b\}^* \text{ e } \#a \text{ seja divisível por } 3\}$.
- (b) $L = \{w \mid w \in \{a, b, c\}^* \text{ e } \#a + \#b \text{ seja par}\}$.
- (c) $L = \{w \mid w \in \{a, b, c\}^* \text{ e } \#a \text{ e } \#b \text{ seja par}\}$.
- (d) $L = \{w \mid w \in \{a, b, c\}^* \text{ e } \#a \text{ ou } \#b \text{ seja par}\}$.
- (e) $L = \{w \mid w \in \{a, b\}^* \text{ e } \#a + \#b \text{ seja múltiplo de } 3\}$.
- (f) $L = \{w \mid w \in \{a, b\}^* \text{ e } w \text{ termina com a sequência } abb\}$
- (g) $L = \{w \mid w \in \{a, b\}^* \text{ e } w \text{ possui pelo menos } 3 \text{ b's consecutivos}\}$
- (h) $L = \{a^i b^j c^k \mid \text{com } i, j, k \geq 0\}$.
- (i) $L = \{a^n b^m c^k \mid \text{com } n \geq 0, m \geq 1, k \geq 2\}$.
- (j) $L = \{a^n (b, c)^* \mid n + \#b \text{ seja múltiplo de } 3\}$
- (k) $L = \{(a, b)^* c^n \mid \#a + n \text{ seja múltiplo de } 3 \text{ e } \#b \text{ seja par}\} \rightarrow \text{Foto 27/03}$



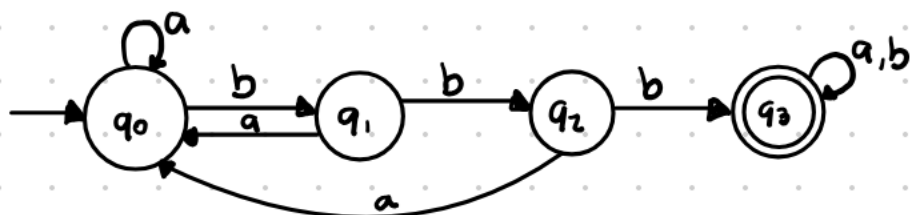
(e) $L = \{w \mid w \in \{a, b\}^* \text{ e } \#a + \#b \text{ seja múltiplo de } 3\}$.



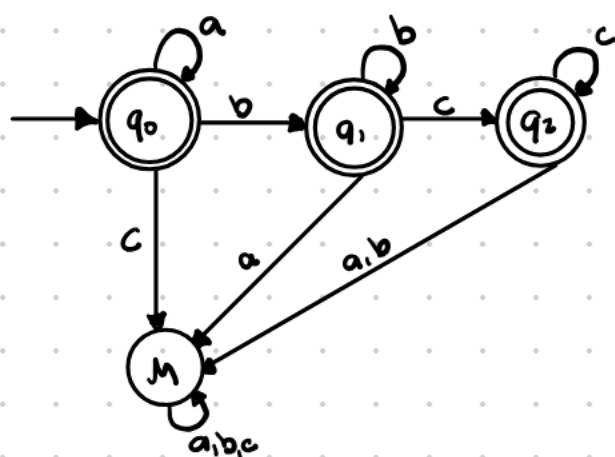
(f) $L = \{w \mid w \in \{a, b\}^* \text{ e } w \text{ termina com a sequência } abb\}$



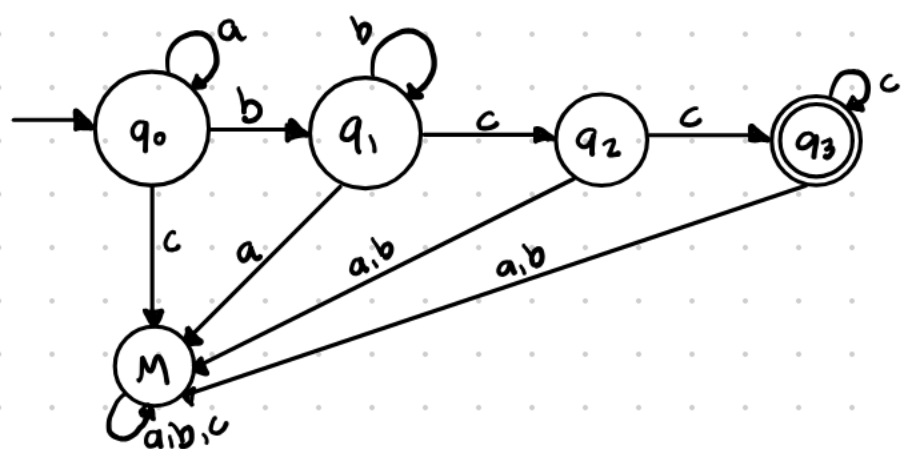
(g) $L = \{w \mid w \in \{a, b\}^* \text{ e } w \text{ possui pelo menos 3 b's consecutivos}\}$



(h) $L = \{a^i b^j c^k \mid \text{com } i, j, k \geq 0\}$.

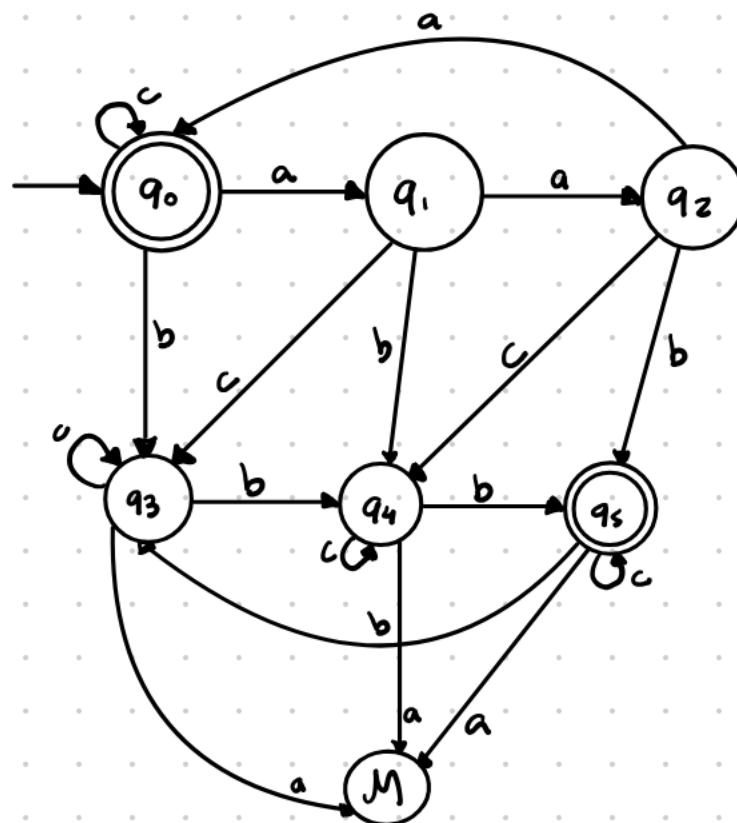


(i) $L = \{a^n b^m c^k \mid \text{com } n \geq 0, m \geq 1, k \geq 2\}$.

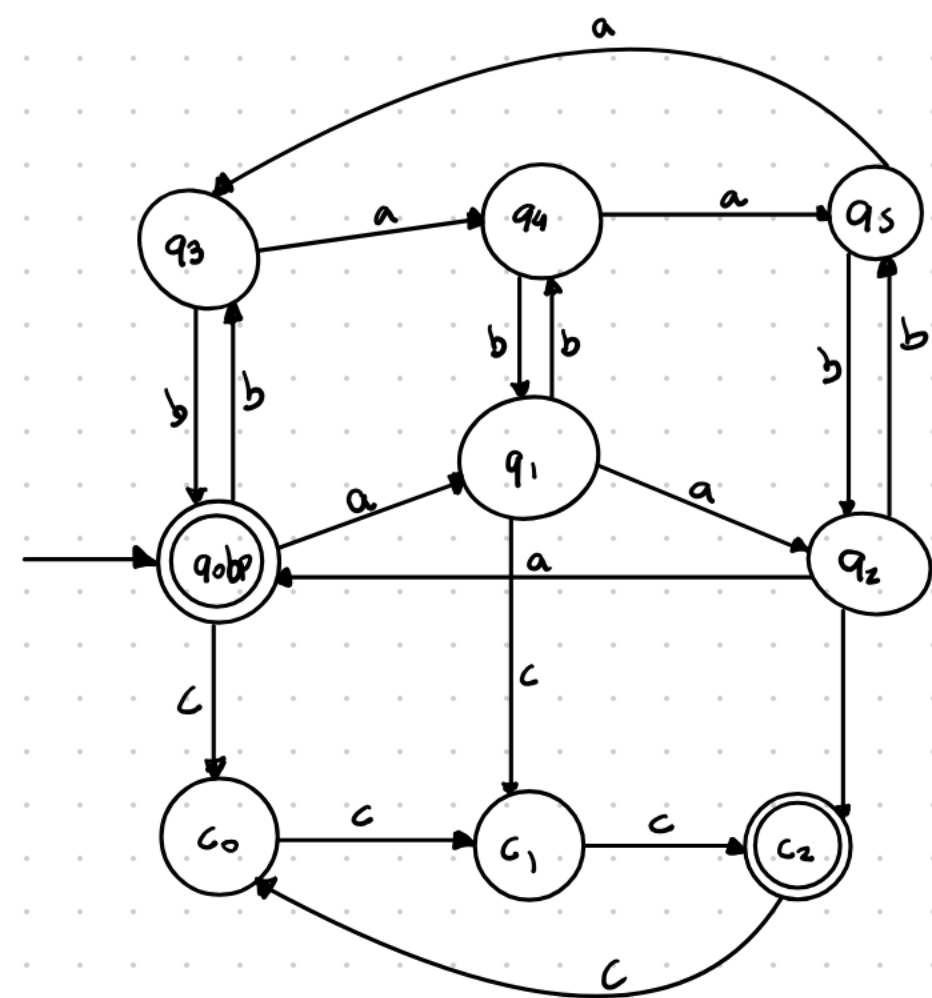


(j) $L = \{a^n (b, c)^* \mid n + \#b \text{ seja múltiplo de } 3\}$

$$A^*(b, c)^* \cap \#A + \#b \equiv 0 \pmod{3}$$



(k) $L = \{(a, b)^* c^n \mid \#a + n \text{ seja múltiplo de } 3 \text{ e } \#b \text{ seja par} \}$



estrutura + cont 1 + cont 2